10. Collection Runner Execution

In API testing using Postman, the **Collection Runner** is used to execute multiple API requests in a sequence automatically. It helps simulate a complete end-to-end workflow of an application without manually running each request.

In this project using Fake Store API, Collection Runner plays a very important role because we test real e-commerce flow like Login → Products → Cart → Users.

10.1 What is Collection Runner?

Collection Runner is a Postman feature that allows you to:

- Run a full collection or folder at once
- Execute APIs in sequence
- Run multiple iterations
- Use test data files (JSON/CSV)
- Automate API execution flow

☞ It is like a **mini automation engine inside Postman**.

10.2 Why Collection Runner is used?

We use Collection Runner for:

- End-to-end API testing
- Batch execution of APIs
- Regression testing
- Data-driven testing
- Verifying workflow dependencies
- Reducing manual effort

10.3 How Collection Runner works in this project

◆ Step-by-step flow:

In our Fake Store API project:

Step 1: Authentication runs first

- Login API executes
- Token is generated
- Stored in environment variable (`authToken`)

Step 2: Product APIs execute

- Get All Products
- Get Single Product
- Add Product
- Update Product
- Delete Product

☞ These APIs may use token or product IDs dynamically.

Step 3: Cart APIs execute

- Add to Cart
- Update Cart
- Get Cart Details
- Delete Cart

Step 4: User APIs execute

- Get all users
- Get single user

10.4 How to Run Collection Runner (Steps)

◆ Step 1:

Open Postman → Click on **Collection**

◆ Step 2:

Click **“Run Collection”**

◆ Step 3:

Select:

- Collection name
- Environment (QA / Dev)

◆ **Step 4:**

Configure settings:

- Iterations (1, 5, 10 etc.)
- Delay (optional)
- Data file (CSV/JSON if required)

◆ **Step 5:**

Click **Run**

10.5 Features Used in Collection Runner

◆ 1. Sequential Execution

- APIs run one after another
- Maintains dependency order

◆ 2. Iterations

- Run same test multiple times
- Example: 5 iterations → same API runs 5 times

◆ 3. Data-driven Testing

- Use external files like CSV or JSON
- Example:

```
[
  { "productId": 1 },
  { "productId": 2 }
]
```

◆ 4. Environment Support

- Uses variables like:
  - {{baseURL}}
  - {{authToken}}

◆ 5. Test Execution Summary

After execution, Postman shows:

- Total requests run
- Passed tests
- Failed tests
- Response time
- Errors

10.6 Example Real Flow in Project

In our project:

1. Login API → generates token
2. Store token in environment
3. Get Products API → fetch data
4. Add Product API → create product
5. Add to Cart API → simulate purchase
6. Get Cart API → validate cart contents

☞ This simulates real e-commerce user behavior.

10.7 Advantages of Collection Runner

- No manual execution needed
- Saves time
- Easy regression testing
- Supports automation basics
- Helps in debugging API flow
- Useful before CI/CD execution

10.8 Limitations

- No advanced reporting (compared to Newman)
- Limited logging
- Not suitable for large CI pipelines
- UI-based (manual start required)

✅ Final Interview Summary

In this project, I used Postman Collection Runner to execute the entire Fake Store API test suite in a sequential and automated manner. It allowed me to run authentication, product, cart, and user APIs together, validate end-to-end workflows, and perform regression testing efficiently. I also used environment variables and data-driven testing to simulate real-world API behavior, which improved test coverage and reduced manual effort.
